Light

GCSE Computer Science - Networks Basics (Set 3)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

Which statement about WANs is correct?

- ○ **A.** WANs typically use infrastructure provided by telecommunication companies
- ○ **B.** WANs are always owned by a single organisation
- ○ **C.** WANs can only connect two computers
- ○ **D.** WANs do not require any routing equipment

Q2 (1 mark)

Why do LANs typically have faster data transfer speeds than WANs?

- ○ **A.** LANs cover shorter distances with dedicated private connections
- ○ **B.** LANs use more expensive computers
- ○ **C.** LANs have more users connected
- ○ **D.** LANs do not use any protocols

Q3 (1 mark)

In a client-server network, what is the role of a client?

- ○ **A.** To store all files for the network
- ○ **B.** To provide services to other computers
- ○ **C.** To request services and resources from the server
- ○ **D.** To manage user passwords for the network

Q4 (1 mark)

What is a disadvantage of peer-to-peer networks?

- ○ **A.** They require expensive server hardware
- ○ **B.** Security and backup management becomes difficult as the network grows
- ○ **C.** They are difficult to set up for small networks
- ○ **D.** They cannot share files between computers

Q5 (1 mark)

What is the key feature of a mesh network topology?

- ○ **A.** All devices connect to a single central hub
- ○ **B.** Devices are connected in a ring formation
- ○ **C.** Devices have multiple connections to other devices, providing redundant paths
- ○ **D.** All devices share a single communication cable

Q6 (1 mark)

What is the main disadvantage of a star network topology?

- ○ **A.** The central hub or switch is a single point of failure
- ○ **B.** Individual device failures affect the entire network
- ○ **C.** It requires more cabling than mesh topology
- ○ **D.** Devices cannot be easily added to the network

Q7 (1 mark)

Which type of transmission media uses light signals to transmit data?

- ○ **A.** Fibre optic cable
- ○ **B.** Copper Ethernet cable
- ○ **C.** Wireless radio waves

- ○ **D.** Coaxial cable

Q8 (1 mark)

What does Wi-Fi use to transmit data?

- ○ **A.** Light signals through fibre optic cables
- ○ **B.** Electrical signals through copper wires
- ○ **C.** Radio waves through the air
- ○ **D.** Sound waves through speakers

Q9 (1 mark)

What is a potential disadvantage of using cloud storage?

- ○ **A.** Files can be accessed from multiple locations
- ○ **B.** Access to data depends on having an Internet connection
- ○ **C.** The cloud provider handles hardware maintenance
- ○ **D.** Storage can be easily scaled up when needed

Q10 (1 mark)

Why do different network protocols exist for different purposes?

- ○ **A.** Because each type of data transmission has specific requirements that need specialised rules

- ○ **B.** To make networks more confusing and difficult to use
- ○ **C.** To increase the cost of network equipment
- ○ **D.** Because older protocols have all been replaced by one universal protocol

Q11 (1 mark)

What is the purpose of an IP address?

- ○ **A.** To identify the manufacturer of a network device
- ○ **B.** To store the device's password
- ○ **C.** To encrypt data being transmitted
- ○ **D.** To uniquely identify a device's location on a network

Q12 (1 mark)

When is a MAC address assigned to a network device?

- ○ **A.** When the device connects to a new network
- ○ **B.** When the user creates an account
- ○ **C.** During manufacturing, before the device is sold
- ○ **D.** When the device is restarted

Q13 (2 marks)

A network technician needs to connect computers within an office so they can share files, and also connect the office network to the Internet. Which devices are needed for each task?

- ○ **A.** A switch for both tasks
- ○ **B.** A router for both tasks
- ○ **C.** A switch to connect computers internally, and a router to connect to the Internet
- ○ **D.** A wireless access point for both tasks

Q14 (2 marks)

A company's DNS server stops working. Users report they cannot access websites by typing domain names, but can still access the same websites using IP addresses directly.
What does this confirm about DNS?

- ○ **A.** DNS translates domain names to IP addresses, and without it, names cannot be resolved
- ○ **B.** DNS is responsible for encrypting all web traffic
- ○ **C.** DNS stores all website content locally
- ○ **D.** DNS provides the Internet connection itself

Q15 (2 marks)

A company sets up a dedicated file server on their network.
What is the main purpose of this server?

- ○ **A.** To connect the network to the Internet
- ○ **B.** To provide wireless network access

- ○ **C.** To convert domain names to IP addresses
- ○ **D.** To store and manage files that users across the network can access

Q16 (2 marks)

A home network has a 100 Mbps Internet connection. The family has 10 devices all streaming video simultaneously.
Why might each device experience slow streaming?

- ○ **A.** The router has run out of storage space
- ○ **B.** The streaming services have stopped working
- ○ **C.** The devices are too far apart from each other
- ○ **D.** The 100 Mbps bandwidth is being shared across all 10 devices, reducing each device's available speed

Q17 (2 marks)

Why is encryption important when using public Wi-Fi networks?

- ○ **A.** It makes the Wi-Fi signal stronger
- ○ **B.** It automatically connects you to the fastest network
- ○ **C.** It increases the speed of your Internet connection
- ○ **D.** It prevents other users on the network from reading your transmitted data

Q18 (2 marks)

A web developer needs to upload website files to a web server, and also wants users to view the finished website in their browsers.
Which protocols are typically used for each task?

- ○ **A.** HTTP for uploading files, FTP for viewing websites
- ○ **B.** DNS for uploading files, HTTP for viewing websites
- ○ **C.** SMTP for both tasks
- ○ **D.** FTP for uploading files, HTTP for viewing websites

Q19 (2 marks)

An IPv4 address has 32 bits, while an IPv6 address has 128 bits.
How many times more addresses can IPv6 provide compared to IPv4?

- ○ **A.** 4 times more addresses
- ○ **B.** 96 times more addresses
- ○ **C.** Approximately 2^96 times more addresses (a vastly larger number)
- ○ **D.** The same number of addresses

Q20 (2 marks)

A network switch receives a data frame and needs to deliver it to the correct device.
Which address does the switch use to determine where to send the frame?

- ○ **A.** The destination MAC address

- ○ **B.** The sender's IP address
- ○ **C.** The website's domain name
- ○ **D.** The sender's email address

Q21 (2 marks)

A company's web server receives a request from a customer's browser for their home page.
What does the web server do?

- ○ **A.** It sends the customer's browser to a different website
- ○ **B.** It processes the request and sends back the web page files
- ○ **C.** It stores the customer's personal information
- ○ **D.** It blocks the request for security reasons

Q22 (2 marks)

When sending an email, the user doesn't need to know how data is physically transmitted through cables or radio waves.
Which benefit of protocol layers does this demonstrate?

- ○ **A.** Encryption
- ○ **B.** Compression
- ○ **C.** Abstraction - hiding complexity from users and applications
- ○ **D.** Authentication

Q23 (2 marks)

Which connection type is generally considered more physically secure against eavesdropping?

- ○ **A.** Wireless, because the signal spreads in all directions
- ○ **B.** Wired, because an attacker would need physical access to the cable to intercept data
- ○ **C.** Both are equally secure against eavesdropping
- ○ **D.** Neither can be made secure

Q24 (3 marks)

A small medical clinic needs to set up a network with the following requirements:

- 8 desktop computers for staff
- Reliable connections that won't affect other computers if one fails
- A central location for backups and patient records
- Internet access for all computers

Which network configuration best meets these requirements?

- ○ **A.** Peer-to-peer network with mesh topology
- ○ **B.** Client-server network with star topology, using a switch, server, and router
- ○ **C.** Peer-to-peer network with each computer connecting directly to the Internet
- ○ **D.** Client-server network with bus topology

Q25 (3 marks)

Maya composes an email on her laptop and sends it to her colleague Tom, who reads it on his smartphone.

Which protocols are involved in this complete email journey?

- ○ **A.** HTTP only - to send and receive the email
- ○ **B.** TCP/IP only - no other protocols needed
- ○ **C.** FTP to transfer the email file, then DNS to deliver it
- ○ **D.** SMTP to send Maya's email to the servers, then IMAP or POP for Tom to retrieve it from his server

Q26 (3 marks)

A photography business is deciding between storing client photos on a local server or using cloud storage. They work in a rural area with unreliable Internet.
Which storage option is more appropriate, and why?

- ○ **A.** Cloud storage, because it provides automatic backups
- ○ **B.** Local server, because it doesn't depend on Internet connectivity for daily access to files
- ○ **C.** Cloud storage, because it's always faster than local storage
- ○ **D.** Local server, because cloud storage cannot handle large image files

Q27 (3 marks)

When a user accesses their online banking website using HTTPS, the data is encrypted.
At which point in the transmission is the data encrypted and decrypted?

- ○ **A.** Encrypted at the router and decrypted at the ISP
- ○ **B.** Encrypted in the user's browser before transmission, and

decrypted at the bank's server after receipt

○ **C.** Encrypted by the DNS server and decrypted by the web server

○ **D.** Encrypted at the user's computer and never decrypted

Q28 (3 marks)

A network administrator notices a device is connected to the network but cannot access the Internet. The device has a MAC address but no IP address assigned.
What is the most likely cause and solution?

○ **A.** The MAC address is invalid; replace the network card

○ **B.** The device needs a new MAC address; contact the manufacturer

○ **C.** The DNS server is down; restart the DNS service

○ **D.** The DHCP server isn't assigning an IP address; check DHCP server functionality or configure a static IP

Q29 (3 marks)

A user reports they can ping a web server's IP address successfully, but cannot view the website in their browser.
What does this tell us about which network layers are working and where the problem likely exists?

○ **A.** All layers are failing; the network is completely down

○ **B.** Lower layers (physical, network) are working; the problem is likely at the application layer (web server software or browser)

○ **C.** The physical cable is broken

○ **D.** The problem is definitely caused by a faulty router

Q30 (3 marks)

A warehouse company needs to set up a network for:

- Fixed workstations at loading bays that process large shipping documents
- Handheld barcode scanners used by workers moving throughout the warehouse

Which network design best meets both needs?

○ **A.** All wireless network - simplest to install throughout the warehouse

○ **B.** All wired network - most reliable for all devices

○ **C.** Wired Ethernet for fixed workstations, Wi-Fi coverage for mobile barcode scanners

○ **D.** Bluetooth connections for everything

Total: **55** marks • Time: **30 min**

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